Exercise path

Start with exercises 1 and 2. Then choose exercise 3 or 4 without help. Exercises 5 to 7 are a bank of transfer, investigation or programming tasks: the teacher assigns one. Exercise 8 is revisited later.

Exercise 7.1 · Constructing a line

The plane is equipped with an orthonormal coordinate system. The line d passes through $A(-1; 3)$ and has the vector $\vec{n} = (2; 5)$ as normal vector.

1. Write the condition for a point $M(x; y)$ to d using a scalar product. Hence deduce a Cartesian equation of d.
2. Give a non-zero direction vector of d, then the coordinates of a point of d distinct from A.
3. Check your point by substituting into the equation.

Exercise 7.2 · Calculating a projection

In an orthonormal coordinate system, we are given the point $P(4; 3)$ and the line d with equation $2x - y = 0$. Let H the orthogonal projection of P on d.

1. Determine the coordinates of H by expressing the two conditions: H belongs to d and $\overrightarrow{PH}$ is normal to d.
2. Check these two conditions separately with the coordinates obtained.
3. Calculate the distance from P to the line d in exact form.

Exercise 7.3 · Recognising a circle

Consider the set of points $M(x; y)$ of the orthonormal plane whose coordinates satisfy

$$x^2 + y^2 - 6x + 4y - 12 = 0.$$

1. Show that this set is a circle and specify its centre and radius.
2. Determine all its intersection points with the line with equation $y = -2$.
3. Interpret geometrically the position of these two points on the circle.

Exercise 7.4 · Perpendicular bisector

In an orthonormal coordinate system, we are given $A(-1; 1)$ and $B(3; 3)$. We seek an equation of the perpendicular bisector of $[AB]$.

1. Determine this equation using the geometric properties of the perpendicular bisector. Explain the choice of a point and a normal vector.
2. Verify that the midpoint of $[AB]$ satisfies the equation obtained.

Exercise 7.5 · The best access

A straight road is represented by the line d with equation $y = 2x$ in an orthonormal coordinate system whose unit of length is 1 km. A building is located at the point $P(4; 3)$. We want to construct a straight access road linking the building to a point on the road.

1. Determine the point on the road that makes this access as short as possible. Calculate the exact length of access.
2. Justify that your choice is optimal among all points on the road, by comparing distances using a right-angled triangle. A single observation on a drawing is not enough.

Exercise 7.6 · Parameter and Set

For a real parameter k, we define $\mathcal{E}_k$ as the set of points $M(x; y)$ whose coordinates satisfy

$$x^2 + y^2 - 2x + 4y + k = 0.$$

1. Determine the type of $\mathcal{E}_k$ depending on the value of k. Organise the answer into disjoint cases.
2. When this set is a circle, specify its centre and radius; in other cases, describe exactly the set obtained.

Exercise 7.7 · A parabola as a locus

Optional extension. In an orthonormal coordinate system, fix the point $F(0; 2)$. We look for the points $M(x; y)$ whose distance to F is equal to the distance to the x-axis.

1. Express the square of each of the two distances in terms of x and y.
2. Deduce an equation of the required set. Check the converse to justify that this equation describes exactly all the required points.

Exercise 7.8 · Delayed review

In an orthonormal coordinate system, $\mathcal{C}$ is the circle with diameter [AB], with $A(0; 0)$ and $B(4; 0)$. Consider the line d with equation $x = 5$.

1. Determine whether $\mathcal{C}$ and d have a common point, by solving an equation.
2. Confirm the answer by comparing the distance from the centre of the circle to the line with the radius of the circle.

Independent work. Do only the task assigned at the end of the session, as set in the dated scheme of work: 10-20 min maximum. The exercises depend on the concepts already studied; some questions are therefore postponed. Revisit exercise 8 one week after the chapter (10 min). If the planned time is exceeded, note the difficulty and bring your attempts; do not try to do the whole reserve.

To be done in the CQFD textbook

These exercises are done in your textbook. The scheme of work gives the session; they replace the homework planned for that day (10 to 20 minutes in total).

CQFD, exercise 18 p. 279, questions a, b, c and d. Read a non-zero normal vector for each equation, without solving for y.

CQFD, exercise 21 p. 279, questions a and b. Write the condition of orthogonality to the normal, then check that the given point lies on the line.